

STUDENT GRADE PREDICTOR

SUMMER INTERNSHIP REPORT

Submitted by

S.AKSHARA (VTU27739)

In partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

**VEL TECH RANGARAJAN DR.SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY-AVADI CHENNAI 600062.**



JUNE 2026

BONAFIDE CERTIFICATE

Certified that this internship report titled "STUDENT GRADE PREDICTOR" is the bonafide work of S.AKSHARA VTU27739 who carried out the internship at "INNOLIFT VENTURES PRIVATE LIMITED" from 01.06.2026 to 20.06.2026 (20 days).

ACKNOWLEDGEMENT

The Internship commenced from 01-06-2026 to 20-06-2026. I have attended the Internship in DOMAIN NAME conducted by INNOLIFT VENTURES PRIVATE LIMITED located in Chennai, Tamilnadu.

INNOLIFT VENTURES PRIVATE LIMITED is a forward-thinking technology company involved in Web application development, web solutions, and AI/ML innovations. It is a private company incorporated and registered with the Registrar of Companies, Chennai. The organization is actively engaged in delivering cutting-edge digital products and services.

I am delighted to extend my sincere appreciation to the significant individuals who have offered their crucial help, inspiration, and unwavering moral support in various ways during my internship. I consider myself extremely fortunate to have received such invaluable guidance and assistance throughout the completion of my internship.

I sincerely thank Dr. SRIVEL and Mr. GOKULAKRISHNAN S, DIRECTORS OF INNOLIFT VENTURES PRIVATE LIMITED, for giving me the opportunity to work on a real-time project and for their constant encouragement and mentorship.

I would also like to extend my heartfelt thanks to the trainers JAYASRI S., M R AKHIL, BALAKRISHNAN J., and VENKATESH K for their dedicated training, technical guidance, and constant support throughout the internship. Their knowledge, practical insights, and willingness to share their expertise greatly contributed to my skill development and understanding of industry practices.

I am truly grateful for their encouragement, valuable suggestions, and unwavering support, which made this internship a meaningful and enriching experience. Thank you all for believing in my potential and contributing to my professional development.

I would like to thank my family and friends, whose unwavering support and encouragement have been instrumental in the completion of this work.

S.AKSHARA

PREFACE

The "STUDENT GRADE PREDICTOR" Machine Learning in Education Project Report encapsulates the practical application of modern Web technologies in building a scalable and user-centric educational platform. This project, undertaken as part of my Summer Internship, highlights the end-to-end development of a cross-platform web application using Python, Pandas, NumPy, Scikit-Learn, Ridge Regression, Flask, HTML, CSS, SQLite, GitHub, and Render, focusing on enhancing digital learning accessibility.

In today's digitally-driven education ecosystem, there is an ever-increasing demand for robust, intuitive, and feature-rich Web applications that cater to both learners and educators. This system was conceptualized to meet those needs, offering modules such as user authentication, course access, quizzes, and progress tracking, all within a responsive and visually engaging user interface.

The development process included dataset collection, data cleaning, feature selection, exploratory data analysis, model training, model evaluation, prediction system development, database integration, user interface design, testing, optimization, version control, cloud deployment, performance validation, result visualization, and live application hosting. The project utilized Python, Flask, HTML, CSS, JavaScript, SQLite, Pandas, NumPy, Scikit-learn, GitHub, VS Code, and Render to build a complete end-to-end machine learning solution for academic performance prediction.

This project was successfully completed during a valuable internship at INNOLIFT VENTURES PRIVATE LIMITED, Chennai. Under the mentorship of experienced developers and project leads, I received hands-on experience in Web app development, applying theoretical concepts to real-world challenges. The internship provided not just technical knowledge but also an understanding of Team Collaboration, Agile Development Methodologies, and product deployment.

Through this journey, I developed significant skills in full-stack Web development, debugging, and user-centered design thinking, all of which have prepared me for future endeavors in the software development field.

COMPANY PROFILE

INNOLIFT VENTURES PRIVATE LIMITED is a private limited company headquartered in Chennai, Tamil Nadu, India. The company was incorporated with a vision to innovate and uplift digital transformation through modern technologies. It is registered with the Registrar of Companies (ROC), Chennai, and classified as a non-government IT company.

Directors of Innolift Ventures Private Limited are Mr. GOKULAKRISHNAN and Dr. SRIVEL.

Innolift Ventures Private Limited is a private company located in Chennai, Chennai District, Tamil Nadu, India.

INNOLIFT VENTURES PRIVATE LIMITED provides internship and other computer related courses which are suitable for computer science and engineering students. They train the students to gain knowledge and skills about their domain.

The internship program was conducted under the guidance of experienced trainers, namely Ms. Jayasri S., Mr. Akhil M R., Mr. Balakrishnan J., and Mr. Venkatesh K. Their expertise, mentorship, and continuous support helped students develop a strong understanding of their domain and improve their technical competencies. The trainers provided practical knowledge, industry insights, and valuable guidance, enabling participants to enhance their problem-solving abilities and professional skills.

Operating in the Information Technology sector, INNOLIFT VENTURES specializes in various computer-related activities, including website maintenance and the creation of multimedia presentations for other firms.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
1	INTRODUCTION TO MACHINE LEARNING IN EDUCATION	1
	1.1 What is Machine Learning?	1
	1.2 Types of Machine Learning	1
	1.3 Machine Learning in Education	2
	1.4 Student Grade Prediction System	2
	1.5 Importance of Student Performance Prediction	3
2	TOOLS AND FRAMEWORK	4
	2.1 Overview	4
	2.2 Programming Language – Python	4
	2.3 Data Analysis Libraries – Pandas and NumPy	4
	2.4 Machine Learning Framework – Scikit-Learn	5
	2.5 Database Management – SQLite	5
	2.6 Web Application Framework – Flask	6
	2.7 Development Environment and Version Control – VS Code and GitHub	6
	2.8 Deployment Tools – Render	7
3	COMPONENTS	7
	3.1 Frontend Components	8
	3.2 Backend Components	9
	3.3 Database Components	9
	3.4 Auxiliary Components	10
	3.5 Role of Each File in the Codebase	10
4	MACHINE LEARNING WORKFLOW	12
	4.1 Understanding Student Performance Prediction	12
	4.2 Dataset Features and Input Variables	12
	4.3 Data Preprocessing Techniques	13
	4.4 Feature Selection and Engineering	13
	4.5 Model Training and Evaluation	14
	4.6 Prediction Generation and Results	15
5	RULES AND EXAMPLE	16
	5.1 Coding Standards in Python	16
	5.2 Machine Learning Development Standards	16
	5.3 Data Preprocessing and Validation Rules	17
	5.4 User Interface Design Guidelines	18
	5.5 Model Performance Optimization	18

	5.6 Example: Student Grade Prediction Process	19
	5.7 Example: Prediction History Storage Using SQLite	19
	5.8 Example: Flask Web Application Integration	20
6	PROJECT VSCODE WEB DEVELOPMENT	21
	6.1 Project Overview	21
	6.2 Objective of the Application	21
	6.3 Project Scope	21
	6.4 Key Features	22
	6.5 User Roles and Flow	22
	6.6 Tools Used for Development	23
	6.7 Screenshots and Descriptions	24
	6.8 Challenges Faced	26
	6.9 Learning Outcomes	26
7	CONCLUSION	27
	7.1 Summary of Learning Outcomes	27
	7.2 Industry Exposure	27
	7.3 Real-Time Application Value	28
	7.4 Personal Experience	28

1. INTRODUCTION TO MACHINE LEARNING IN EDUCATION

1.1 What is Machine Learning

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions without being explicitly programmed. It uses algorithms and statistical techniques to identify patterns, analyze data, and improve prediction accuracy over time. Machine Learning is widely used in various fields such as healthcare, finance, business, education, and technology. In recent years, Machine Learning has become an important tool for solving real-world problems by providing intelligent solutions based on data-driven insights. It helps organizations automate tasks, improve decision-making, and increase efficiency.

1.2 Types of Machine Learning

Machine Learning is mainly categorized into the following types:

a) Supervised Learning

- Uses labeled datasets for training.
- Learns the relationship between input and output variables.
- Common algorithms include Linear Regression, Ridge Regression, and Decision Trees.

b) Unsupervised Learning

- Works with unlabeled data.
- Identifies hidden patterns and relationships in datasets.
- Common techniques include Clustering and Association Analysis.

c) Machine Learning in Education

- Student performance prediction
- Personalized learning recommendations
- Attendance analysis
- Academic risk assessment
- Learning behavior analysis

1.3 Machine Learning in Education

Machine Learning plays a significant role in the education sector by analyzing student data and predicting academic performance. It helps educational institutions

identify students who may require additional support and enables data-driven decision-making.

Applications of Machine Learning in education include:

- Student performance prediction
- Personalized learning recommendations
- Attendance analysis
- Academic risk assessment
- Learning behavior analysis

1.4 Student Grade Prediction System

The development of the Student Grade Prediction System involved multiple phases to ensure accurate predictions and a user-friendly application:

1. Data Collection

Gather student academic data from the UCI Student Performance Dataset.

2. Data Preprocessing

Clean the dataset, handle missing values, encode categorical features, and prepare data for analysis.

3. Exploratory Data Analysis (EDA)

Analyze patterns, relationships, and trends using charts, graphs, and statistical methods.

4. Feature Selection

Select the most relevant features affecting student performance, such as study time, failures, absences, and internet access.

5. Model Development

Train Machine Learning models including Linear Regression and Ridge Regression using the prepared dataset.

6. Model Evaluation

Evaluate model performance using metrics such as RMSE, MAE, and R^2 Score to select the best-performing model.

7. Web Application Development

Develop a Flask-based web application with an interactive interface for entering student details and viewing predictions.

8. Deployment and Testing

Deploy the application using Render, test all functionalities, and ensure reliable performance in a live environment.

9. Maintenance and Improvements

Monitor system performance, fix issues, and enhance features based on future requirements.

1.5 Importance of Student Performance Prediction in Education

Student performance prediction has transformed traditional educational analysis by enabling institutions to identify academic trends and support students more effectively. The Student Grade Prediction System is designed as an intelligent tool for students and educators, helping them understand expected academic outcomes based on various educational factors.

Key benefits of Student Performance Prediction:

- Early identification of academically weak students
- Improved academic planning and decision-making
- Data-driven analysis of student performance
- Better guidance and support for students
- Enhanced educational outcomes and learning efficiency

The internship focused on developing this Machine Learning-based prediction system using Python, Scikit-Learn, Flask, and SQLite, addressing both technical challenges and practical educational requirements. The project provided valuable insights into data analysis, predictive modeling, web application development, and real-world deployment.

2. TOOLS AND FRAMEWORK

2.1 Overview

The Student Grade Prediction System was developed using a combination of Machine Learning, Data Analysis, Web Development, Database Management, and Deployment technologies. The selected tools and technologies were chosen based on their efficiency, reliability, ease of implementation, and suitability for predictive analytics. The project utilizes Python for programming, Scikit-Learn for Machine Learning, Flask for web application development, SQLite for database management, and Render for deployment.

2.2 Programming Language – Python

Python is a high-level, interpreted programming language widely used in Data Science, Machine Learning, and Web Development. It provides a simple syntax and extensive libraries that make application development easier and more efficient.

Key Features:

- Easy-to-read and maintain code
- Extensive library support
- Platform independent
- Strong community support
- Suitable for Machine Learning applications

Role in the Project:

- Data preprocessing
- Model development
- Flask application development
- Database integration

2.3 Data Analysis Libraries – Pandas and NumPy

Pandas and NumPy were used for data manipulation and numerical computations.

PANDAS:

Pandas provides powerful data structures for handling datasets efficiently.

Functions Used:

- Reading CSV files
- Data cleaning
- Feature selection
- Data transformation

NUMPY:

NumPy provides support for numerical operations and multidimensional arrays.

2.4 Machine Learning Framework – Scikit-Learn

Scikit-Learn is one of the most popular Machine Learning libraries in Python. It provides tools for model training, evaluation, preprocessing, and hyperparameter tuning.

Algorithms Used:

- Linear Regression
- Ridge Regression

Features:

- Model training and testing
- Cross-validation
- Performance evaluation
- Hyperparameter tuning

Benefits:

- Easy implementation
- High accuracy
- Reliable performance

2.5 Database Management – SQLite

SQLite is a lightweight relational database management system used to store prediction history generated by the application.

Advantages:

- Simple and lightweight
- No separate server required

- Easy integration with Flask
- Efficient data storage

Role in the Project:

- Store prediction records
- Retrieve historical predictions
- Manage user-generated data

2.6 Web Application Framework – Flask

Flask is a lightweight Python web framework used to build the Student Grade Prediction web application.

Key Features:

- Simple architecture
- Easy routing
- Template rendering
- Database integration
- Fast development process

Functions in the Project:

- User interface development
- Form handling
- Prediction processing
- Result display

2.7 Development Environment and Version Control – VS Code and GitHub

Visual Studio Code (VS Code)

VS Code was used as the primary development environment.

Features:

- Integrated terminal
- Python support
- Debugging tools
- Extension support

GitHub

GitHub was used for version control and project management.

Benefits:

- Code backup
- Version tracking
- Collaboration support
- Repository management

2.8 Deployment Platform – Render

Render is a cloud deployment platform used to host the Student Grade Prediction application and generate a live URL.

Deployment Steps:

- Upload project to GitHub
- Connect repository with Render
- Configure build and start commands
- Deploy application

Advantages:

- Free hosting support
- Easy deployment process
- Automatic updates from GitHub
- Public live URL generation

Key Observations from Literature Survey:

- Educational data mining plays an important role in analyzing student performance and learning patterns.
- Various machine learning algorithms have been used in academic prediction systems to improve decision-making.
- Student attendance, study time, previous grades, and failures are commonly identified as significant factors influencing performance.
- Web-based prediction systems provide easy access and improve usability for students and educators.
- Recent studies highlight the effectiveness of predictive analytics in supporting academic planning and student success.

3. COMPONENTS

3.1 Overview of Web App Components

Every Machine Learning application is built using a combination of software components that manage data processing, prediction generation, user interaction, and data storage. In the Student Grade Prediction System, the architecture is divided into frontend, backend, and database layers for better maintainability, scalability, and performance.

3.2 Frontend Components

Frontend refers to the user-facing part of the application where users interact with the system. The frontend of the Student Grade Prediction System was developed using HTML, CSS, and Flask templates.

Key Frontend Elements:

1. Home Page

- Acts as the main entry point of the application.
- Provides project information and navigation options.
- Allows users to access prediction services.

2. Prediction Form

- Collects student-related information.
- Accepts input values such as study time, failures, absences, and internet access.
- Sends data to the backend for processing.

3. Result Page

- Displays predicted student grades.
- Shows academic performance classification.
- Provides user-friendly output presentation.

3.3 Backend Components

Backend components handle the application logic, Machine Learning operations, and data processing. Flask was used as the backend framework.

1. Flask Application

- Handles client requests.
- Manages routing and template rendering.
- Connects frontend with Machine Learning modules.

2. Machine Learning Model

- Ridge Regression model used for prediction.
- Processes user input data.
- Generates predicted student grades.

3. Data Processing Module

- Performs input validation.
- Prepares data for prediction.
- Applies required transformations.

3.4 Database Components

SQLite was used as the database management system for storing prediction history and user-generated records.

Database Features:

Prediction Records

- Stores prediction history.
- Saves generated results.
- Maintains prediction timestamps.

Data Storage

- Stores input values.
- Preserves prediction outputs.
- Supports future analysis.

Data Retrieval

- Retrieves previous prediction records.
- Displays stored results to users.
- Supports historical tracking.

3.5 Auxiliary Components

Apart from the primary modules, the Student Grade Prediction System contains additional supporting components that improve usability and performance.

best_model.pkl	Stores trained Ridge Regression model
scaler.pkl	Stores feature scaling information
features.pkl	Stores selected feature names
metrics.pkl	Stores model evaluation metrics
app.py	Main Flask application
index.html	Prediction input page
result.html	Displays prediction results
predictions.db	SQLite database for prediction history

3.6 Role of Each File in the Codebase

Each file in the project performs a specific function to ensure smooth application execution.

app.py

- Main application file.
- Handles routing and prediction logic.

best_model.pkl

- Contains trained Ridge Regression model.
- Used for generating predictions.

scaler.pkl

- Applies feature scaling.
- Maintains consistency between training and prediction.

features.pkl

- Stores selected model features.
- Ensures proper feature mapping.

metrics.pkl

- Stores RMSE and model evaluation metrics.
- Used for performance analysis.

index.html

- Provides user interface for data entry.
- Collects student information.

result.html

- Displays prediction output.
- Shows final academic score.

predictions.db

- Stores prediction history.
- Maintains database records.

System Workflow:

The system follows a structured workflow starting from data collection and preprocessing to model training, prediction generation, and result display. Important features such as study time, absences, failures, and other academic factors were selected to improve prediction accuracy. Data cleaning, transformation, and encoding techniques were applied to prepare the dataset for machine learning model training. Ridge Regression was implemented and evaluated to predict student grades based on the selected input features. User inputs are processed through the trained model to generate predicted grades and performance insights. The machine learning model was integrated with a Flask web application to provide real-time prediction functionality.

4. MACHINE LEARNING WORKFLOW

4.1 Understanding Student Performance Prediction

In Machine Learning, Student Performance Prediction refers to the process of analyzing academic and behavioral factors to estimate a student's future academic performance. The prediction is generated using historical student data and statistical learning algorithms. In the Student Grade Prediction System, Machine Learning models are trained to identify patterns between input features and final grades.

The prediction model uses various educational attributes such as study time, failures, absences, internet access, and previous academic records. These features are processed and analyzed to generate an estimated academic score.

Machine Learning models act as intelligent prediction engines that learn from historical data and make future predictions. The Ridge Regression algorithm was selected for this project because of its ability to reduce overfitting and improve prediction accuracy.

4.2 Dataset Features and Input Variables

The Student Grade Prediction System uses the UCI Student Performance Dataset. The dataset contains demographic, social, and academic information about students.

Important Features Used:

Study Time	Measures time spent studying
Failures	Number of previous failures
Absences	Total absences of a student
Internet	Internet availability at home
G1	First period grade
G2	Second period grade
G3	Final grade (Target Variable)

These features help the model understand student learning behavior and predict academic performance.

4.3 Data Preprocessing Techniques

Data preprocessing is one of the most important stages in Machine Learning because raw datasets often contain inconsistencies, missing values, and categorical variables that cannot be directly used for model training. In the Student Grade Prediction System, several preprocessing techniques were applied to improve data quality and prediction accuracy.

Preprocessing Operations:

Data Cleaning	Removes unnecessary or inconsistent data
Missing Value Handling	Identifies and processes incomplete records
Categorical Encoding	Converts categorical data into numerical values
Feature Scaling	Standardizes numerical features
Feature Selection	Selects important variables for prediction

The preprocessing stage ensures that the dataset is suitable for Machine Learning algorithms and improves overall model performance.

4.4 Feature Selection and Engineering – Preparing Input Variables

Feature selection is the process of identifying the most relevant variables that influence student academic performance. Selecting important features helps reduce model complexity and improve prediction accuracy.

Selected Features:

Study Time	Measures student learning effort
Failures	Indicates previous academic difficulties
Absences	Represents attendance behavior
Internet Access	Reflects availability of learning resources

Feature engineering was also performed to transform raw data into a format suitable for model training. This process improved the effectiveness of the Ridge Regression model and enhanced prediction quality.

4.5 Model Training and Evaluation

The trained models were evaluated using various performance metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R^2 Score. These metrics helped measure the accuracy and reliability of the predictions generated by the models. Cross-validation techniques were also used to ensure consistent model performance across different subsets of the dataset.

After comparing the results, Ridge Regression achieved better performance and produced more reliable predictions than Linear Regression. Therefore, Ridge Regression was selected as the final model for the Student Grade Prediction System. The trained model was saved as a PKL file and later integrated into the Flask web application for real-time prediction generation.

4.6 Prediction Generation and Results

The trained Ridge Regression model was integrated into a Flask web application to generate student grade predictions.

Prediction Workflow:

- User enters student details.
- Input data is validated.
- Features are preprocessed.
- Ridge Regression model generates prediction.
- Predicted score is displayed.
- Prediction history is stored in SQLite.

The final system successfully predicts student academic performance and demonstrates the practical application of Machine Learning in the education domain. The application was successfully deployed on Render and tested in a live environment. The prediction results indicated that study habits, attendance, and previous academic performance significantly influence student grades. The model demonstrated stable performance and provided useful insights for academic improvement. The developed system achieved the project objective of predicting student grades through an accessible web-based platform.

5. RULES AND EXAMPLE

5.1 Importance of Coding Standards and Best Practices

Following consistent coding standards and development practices is essential for building reliable, maintainable, and scalable Machine Learning applications. During the development of the Student Grade Prediction System, various industry-standard practices were followed to ensure code quality, prediction accuracy, security, and usability.

Special attention was given to:

- Code organization and readability
- Data preprocessing and validation
- Machine Learning model development
- Database management
- User interface consistency
- Error handling and debugging
- Deployment and testing

5.2 Python Coding Standards Followed

The Student Grade Prediction System was developed following standard Python programming guidelines to improve readability and maintainability.

a) Code Structure and Naming

- Files named using meaningful names (app.py, train_model.py)
- Variables named using snake_case (study_time, prediction_score)
- Functions named descriptively (predict_grade(), load_model())
- Code divided into logical modules and files

b) Modular Programming

- Separate files used for training, prediction, and deployment
- Reusable functions created wherever possible
- Reduced code duplication

c) Error Handling

- try-except blocks used for exception handling

- Input validation implemented before prediction
- Error messages displayed for invalid inputs

d) Input Validation

- User inputs validated before processing
- Prevented empty or incorrect values
- Improved system reliability

e) Documentation and Comments

- Important sections documented using comments
- Code organized for easier understanding and debugging

5.3 Data Validation and Security Rules

The Student Grade Prediction System processes user-provided academic information. Therefore, validation and security measures were implemented to ensure reliable predictions and safe data handling.

Validation Rules:

- Required fields must not be empty
- Numeric fields accept only valid numbers
- Invalid values are rejected before prediction
- Input ranges are verified before processing

Database Security:

SQLite was used for storing prediction history. The following measures were followed:

- Secure database connections
- Controlled insertion of prediction records
- Prevention of duplicate or corrupted records
- Proper handling of database exceptions

These validation and security practices help maintain data quality and improve the reliability of the prediction system.

5.4 User Interface Design Guidelines

A simple and user-friendly interface is important for improving user experience and ensuring smooth interaction with the application. During the development of the Student Grade Prediction System, several UI/UX guidelines were followed.

Design Rules Followed:

- Clean and simple page layout
- Consistent spacing and alignment throughout the application
- Easy-to-understand form fields and labels
- Responsive design for different screen sizes
- Clear navigation between pages
- User-friendly prediction result display

These design principles helped create an application that is easy to use and visually appealing.

5.5 Model Performance Optimization

Several optimization techniques were applied to improve the performance and accuracy of the Machine Learning model.

Optimization Techniques:

- Data preprocessing and cleaning
- Feature selection to remove unnecessary variables
- Feature scaling using StandardScaler
- Cross-validation for model evaluation
- Hyperparameter tuning using Ridge Regression

Benefits:

- Improved prediction accuracy
- Reduced overfitting
- Better model generalization
- Faster prediction generation

These optimization techniques helped the model produce reliable and consistent results.

5.6 Example: Student Grade Prediction Process

The Student Grade Prediction process is the core functionality of the application. Users enter academic information through a web form, and the trained Machine Learning model generates a predicted grade.

Features Implemented:

- Input validation
- Machine Learning prediction
- Real-time result generation
- User-friendly result display

5.7 Example: Prediction History Storage Using SQLite

The application stores prediction records using SQLite. This feature helps maintain historical prediction data and allows users to review previous results.

Stored Information:

- Prediction ID
- Date and Time
- Input Features
- Predicted Grade

Benefits:

- Data persistence
- Historical tracking
- Easy record retrieval
- Improved data management

5.8 Example: Flask Web Application Integration

Flask was used to integrate the Machine Learning model with the web application. It handles user requests, prediction generation, and result display.

Features Implemented:

- Form submission handling
- Prediction processing
- Result page rendering.

6. PROJECT – STUDENT GRADE PREDICTION SYSTEM

6.1 Project Overview

The Student Grade Prediction System is a Machine Learning-based web application developed to predict student academic performance using various educational and behavioral factors. The system analyzes student-related information such as study time, absences, failures, and internet access to estimate the final academic grade.

The project combines Machine Learning techniques with web application development to provide an easy-to-use platform for academic performance prediction. The application was developed using Python, Scikit-Learn, Flask, SQLite, HTML, and CSS and was successfully deployed using Render.

6.2 Objective of the Application

The primary objective of the Student Grade Prediction System is to predict student academic performance using Machine Learning algorithms and provide insights that help students and educators make informed decisions.

Specific objectives include:

- Predict student grades accurately using historical academic data.
- Analyze factors affecting student performance.
- Develop a user-friendly web application for prediction generation.
- Store prediction history for future reference.
- Demonstrate the practical application of Machine Learning in education.

6.3 Project Scope

The scope of the Student Grade Prediction System includes the collection, processing, analysis, and prediction of student academic performance.

The system is capable of:

- Accepting student-related input data.
- Processing and validating user inputs.
- Generating academic performance predictions.
- Storing prediction records using SQLite.
- Displaying prediction results through a web interface.

- Supporting future enhancements and model improvements.

The project demonstrates how Machine Learning can be applied to educational analytics and decision-making.

6.4 Key Features of Student Grade Prediction System

The application includes several features that improve functionality and usability.

Major Features:

- Student grade prediction using Ridge Regression.
- Interactive web-based user interface.
- Real-time prediction generation.
- Input validation and error handling.
- Prediction history storage using SQLite.
- Machine Learning model integration.
- Responsive web design.
- Cloud deployment using Render.
- Easy-to-use prediction workflow.
- Reliable and scalable architecture.

6.5 User Roles and Flow

The Student Grade Prediction System follows a simple workflow that allows users to generate predictions efficiently.

User Flow:

- User accesses the application.
- Student details are entered into the prediction form.
- Input values are validated.
- Data is processed and transformed.
- Machine Learning model generates prediction.
- Predicted grade is displayed.
- Prediction record is stored in SQLite.
- User can view prediction history.

User Role:

- Enter student-related information.
- Generate academic performance predictions.
- View prediction results.
- Access stored prediction history.

6.6 Tools Used for Development

The following technologies were used during the development of the Student Grade Prediction System:

Python	Programming Language
Pandas	Data Analysis
NumPy	Numerical Computation
Scikit-Learn	Machine Learning
Ridge Regression	Prediction Model
Flask	Web Framework
HTML	User Interface
CSS	Styling
SQLite	Database Management
GitHub	Version Control
VS Code	Development Environment
Render	Application Deployment

Each technology was selected based on its efficiency, reliability, and compatibility with the project requirements. The combination of machine learning, web development, database management, and cloud deployment enabled the development of a complete end-to-end application. These tools helped improve development speed, maintainability, scalability, and overall project performance.

6.7 Screenshots and Descriptions

The application is live at: <https://student-gradee-prediction.onrender.com>

Home and Prediction Page:

The home page serves as the main entry point of the GradeIQ application. It features an AI-Powered Student Success Analytics Platform with an Academic Insights

Dashboard showing 395 student records analyzed, an average predicted score of 14.8, a top academic score of 20, and 32 students requiring support. The prediction form collects student details including name, previous academic challenges, available self-study time, social activity level, academic guidance availability, weekly learning commitment, class attendance gaps, wellbeing and energy level, and learning support environment. Users can generate academic forecasts and view prediction history from this page.

About and Prediction History Page:

The About page displays the Model Performance Summary showing a Linear Regression RMSE of 4.21, Ridge Regression RMSE of 4.18, with Ridge selected as the final model. Performance categories are defined as Excellent (Score 16–20), Good (Score 12–15), Average (Score 10–11), and Needs Improvement (Score Below 10). The Prediction History page shows a table of past predictions with student names, date and time, predicted scores, and performance categories.

6.8 Challenges Faced

Several challenges were encountered during the development and deployment of the project.

Technical Challenges:

- Data preprocessing and dataset formatting issues.
- Feature selection and model optimization.
- Managing Machine Learning model files.
- Integrating the trained model with Flask.
- Database connectivity and prediction storage.
- Deployment configuration on Render.
- Resolving dependency and package issues.
- Testing application functionality after deployment.

Solutions Implemented:

- Applied proper preprocessing techniques.
- Compared multiple Machine Learning models.
- Optimized project structure and code organization.
- Configured deployment settings correctly.
- Performed extensive testing and debugging.

6.9 Learning Outcomes

From this project, I gained deep hands-on experience in:

- Machine Learning model development and evaluation.
- Data preprocessing and feature engineering.
- Exploratory Data Analysis (EDA).
- Ridge Regression implementation.
- Python programming and debugging.
- Flask web application development.
- HTML and CSS-based frontend design.
- SQLite database integration.
- GitHub version control and project management.
- Cloud deployment using Render.
- Problem-solving and debugging techniques.
- Real-world software development workflow.

This project significantly improved my technical knowledge and practical skills in Machine Learning, Web Development, Database Management, and Cloud Deployment while providing valuable industry-oriented experience.

7. CONCLUSION

The internship at Innolift Ventures Private Limited, Chennai, has provided invaluable exposure to real-world software development through the successful design and implementation of a Web application titled "STUDENT GRADE PREDICTOR". This internship has enhanced my technical knowledge and practical experience in various ways.

The Student Grade Prediction System was successfully designed, developed, and deployed as a complete machine learning-based web application. The project demonstrated how data-driven techniques can be used to analyze student information and predict academic performance with reasonable accuracy. Through this project, various stages such as data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, web application development, database integration, and cloud deployment were successfully completed.

The implementation of Ridge Regression enabled effective prediction of student grades while maintaining model stability and performance. The developed system not only predicts grades but also provides useful study recommendations, making it beneficial for students to understand and improve their academic performance. The deployment of the project on a live platform further enhanced its usability and accessibility.

This internship provided valuable practical experience in Machine Learning, Web Development, Data Analytics, Version Control, and Cloud Deployment. It strengthened my problem-solving abilities, debugging skills, project management knowledge, and understanding of real-world software development practices. Working on an end-to-end project helped bridge the gap between theoretical concepts and practical implementation.

Overall, the internship has been a highly rewarding learning experience that enhanced both my technical and professional skills. The knowledge gained through this project will serve as a strong foundation for future work in Artificial Intelligence, Machine Learning, and Software Development.

7.1 Summary of Learning Outcomes

During this internship, I gained practical knowledge in Machine Learning, Data Analysis, Flask Web Development, GitHub version control, and cloud deployment. I learned how to work with real-world datasets, build predictive models using Ridge Regression, evaluate model performance, and integrate machine learning.

7.2 Industry Exposure

The internship provided hands-on experience with industry-standard tools and technologies such as Python, Pandas, NumPy, Scikit-learn, Flask, GitHub, VS Code, and Render. I also learned the importance of project planning, documentation, testing, deployment, and presentation in a professional development environment.

7.3 Real-Time Application Value

The developed Student Grade Prediction System demonstrates how machine learning can be applied to analyze student-related factors and predict academic performance. The project provides useful insights and study recommendations, helping students understand areas where improvement is needed and supporting better academic decision-making.

7.4 Personal Experience

This internship significantly improved my technical, analytical, and problem-solving skills. I gained confidence in developing complete end-to-end projects and learned how to overcome real-world challenges during implementation and deployment. The experience has strengthened my interest in Artificial Intelligence and Machine Learning and motivated me to continue learning advanced technologies.

Final Statement

"This internship has not only helped me gain technical expertise in Python, Machine Learning, Flask, GitHub, and Render deployment, but has also empowered me to understand how an idea can be transformed into a scalable real-world solution. I sincerely thank Innolift Ventures Pvt. Ltd. for providing me with this incredible learning opportunity."